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A heterogeneous learner fusion method with supplementary feature for lithium-ion batteries state of health estimation

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ABSTRACT

Accurate estimation of the state of health (SOH) is crucial for ensuring the stable operation of lithium-ion batteries. A new heterogeneous learner fusion SOH estimation method, PCA-SEL-FM, is proposed based on stacking ensemble learning and supplementary feature, in which Gaussian process regression, Support Vector Regression, Extreme Learning Machine, and Long Short-Term Memory Neural Network are selected as four preferred base-learners by numerical experiment. The Extreme Learning Machine is also selected as the preferred meta-learner, while adding supplementary feature to the meta-learner. Those supplementary features are selected from a multi-dimensional health feature by Principal Component Analysis, it makes up for the defect that the meta-learner cannot interact directly with the original data. By comparing with four single data-driven methods and a fusion method without supplementary feature, the numerical results show that the fusion method PCA-SEL-FM has higher accuracy and generalization ability on both large-cycle and small-cycle li-ion battery datasets, with less than 0.005 RMSE and 0.5 % MAPE.

1. Introduction

As an essential part of the new energy sector, lithium-ion batteries have been widely used in aerospace [1], electric vehicles [2], and other fields. Despite the rapid advancements in battery technology, the potential safety hazards associated with li-ion batteries must not be ignored. Therefore, the precise estimation of the state of health (SOH) is crucial for ensuring the safety of li-ion batteries [3].

At present, many scholars have proposed various methods to enhancing the precision of SOH estimation, such as equivalent circuit models (ECM) [4,5], empirical models (EM) [6,7], statistical models (SM) [8,9], data-driven models (DDM) [10], among others. Among these, model-based methods (ECM, EM, and SM) have laid a solid foundation for SOH estimation of li-ion batteries. But the excellent estimation results of model-based methods rely on the quality of the li-ion battery data samples used. In addition, the extraction of health features (HFs) from operational battery data is crucial for data-driven models. Even with valid HFS, it is often difficult to achieve optimal prediction accuracy using a single DDM.

Therefore, fusion methods integrating multiple models have provided important research ideas to the accurate estimation of SOH [11,12]. Based on ECM and data-driven fusion strategies, an improved anti-noise adaptive long short-term memory (ANA-LSTM) neural network with high-robustness feature extraction and optimal parameter characterization is proposed for accurate remaining useful life (RUL) prediction [13]. These methods combine EM and DDM, not only simulating battery degradation trend, but also leveraging machine learning (ML) methods to describe degradation differences. Furthermore, many research results have been produced in the development of multi-DDM fusion. For example, a SOH estimation method with error compensation was developed where the Autoregressive (AR) model was used to predict long-term trend of battery degradation and the Relevance Vector Machine (RVM) learned error sequence [14]. A linearly weighted combination of the autoregressive integrated moving average (ARIMA) model and LSTM was used to predict battery capacity [15]. An improved robust multi-time scale singular filtering-GPR-LSTM modeling method was proposed for the remaining capacity estimation [16]. Overall, these multi-DDM fusion methods have demonstrated promising prediction results. It is worth mentioning that combining multiple-ML methods into a whole have become a promising approach for SOH estimation [17]. However, in case of diverse of li-ion battery data and insufficient data processing capabilities, the fusion of multi-ML may not effectively leverage available information and faces challenges such as limited generalization ability and susceptibility to overfitting.

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Table 1
The test parameters of three battery datasets.

Dataset	Battery	Nominal Capacity	Charging strategy	Discharging strategy	Discharge Cut-off voltage
A	Cell1	740mAh	1C CC	1C CC	2.7 V
	Cell3				2.7 V
	Cell7				2.7 V
	Cell8				2.7 V
B	B0005	2 Ah	1C CC-CV	1C CC	2.7 V
	B0006				2.5 V
	B0007				2.2 V
C	batch21	1.1 Ah	5.4C (80 %)-5.4C	4C CC	2 V
	batch1		5C (67 %)-4C-1C		2 V

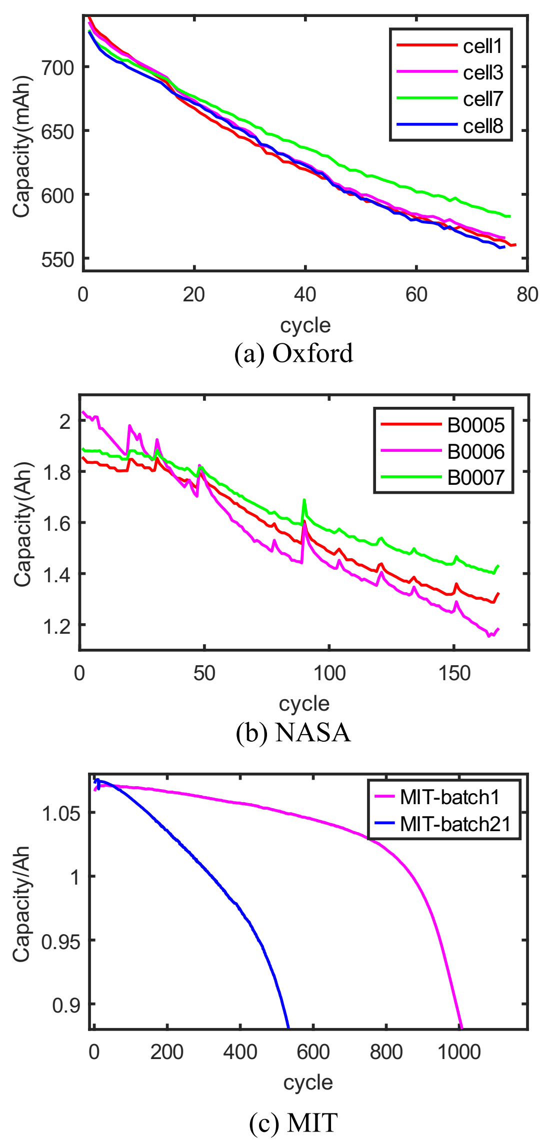


Fig. 1. Li-ion batteries capacity degradation curves.

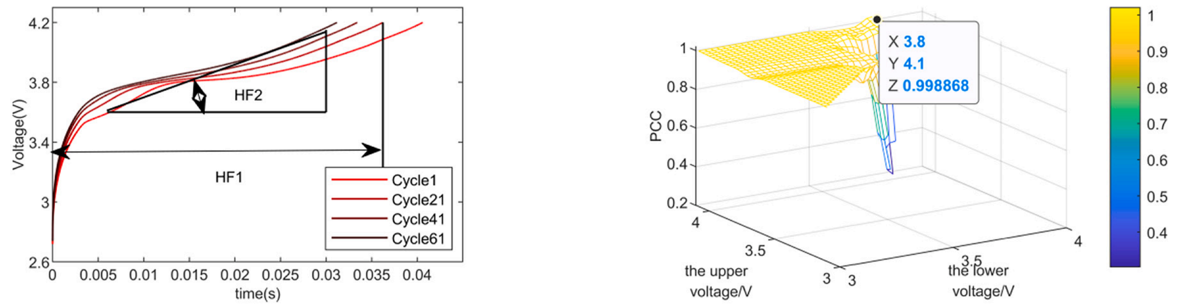
To address these issues, ensemble learning (EL) mechanism has been developed [18]. EL integrates multiple ML algorithms using specific fusion mechanisms to achieve more accurate estimation and it has been widely used in SOH estimation [19–22]. For example, the error rate of each ML algorithm was calculated to determine the corresponding weight, and the integration of multiple support vector regression (SVR) models was realized [19]. Designing a weighted strategy based on predicting uncertainty, a multiple-GPR model was proposed for SOH estimation [20]. By integrating multiple sequential parallel extreme learning machine (ELM), the RUL of electronic devices was predicted and the average root mean square error (RMSE) of the model is as low as 0.78 % on a public dataset [21]. Even if their excellent performance, these existing works often rely on fusion models generated by a single learning algorithm [22], which may have limitations such as the challenge of processing large volumes of efficient information and limited generalization ability.

Stacking ensemble learning (SEL) is a heterogeneous ensemble method proposed by Wolpert with high generalization ability [23]. It combines two or more ML algorithms to compensate the limitations of the poor performance of a single algorithm. SEL involves two core concepts: base-learner and meta-learner. The base-learner plays the role of obtaining preliminary estimation results from the original data, and the meta-learner is responsible for obtaining the final prediction results from the results of multiple base-learners. While SEL has been applied in aircraft engine RUL prediction [24] and wind speed prediction [25], there is a dearth of study examining SEL in SOH estimation. A prediction model using stacking ensemble learning algorithm was proposed where the LSTM was used to relearn the weights of SVR models [26]. A stacking ensemble prediction model was constructed for estimating SOH, in which three GPR model of different kernel functions acted as base-learners and linear regression (LR) was the meta-learner, achieving the RMSE lower than 1.0852 % [27]. The base-learners of existing SEL model inherently employed a single learning algorithm, missing out on the benefits of heterogeneous integration. Furthermore, the meta-learner in these models do not directly process the original data, potentially overlooking valuable information and affecting the final effect in turn.

Therefore, a new fusion method that integrates multiple heterogeneous learners and adds supplementary feature is proposed to improve the accuracy of li-ion battery SOH estimation, called PCA-SEL-FM. The main contributions are as follows:

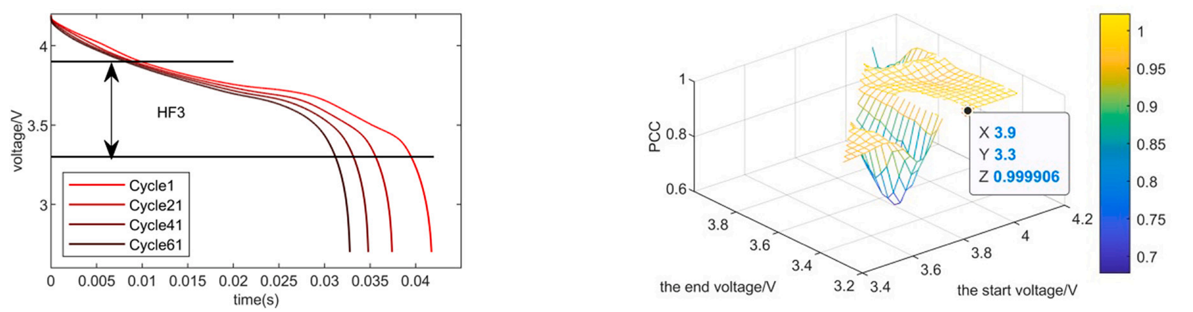
(1) A multi-dimensional HFs based on the charge-discharge voltage data is extracted and optimized by Random Forest (RF) and correlation analysis.
(2) An improved stacking ensemble learning algorithm called PCA-SEL is proposed, The PCA-SEL integrates multiple heterogeneous learners instead of common homogeneous learners, also a supplementary feature selected by Principal Component Analysis is provided as the inputs for the meta-learner. Those supplementary features are selected from multi-dimensional HFs.
(3) Based on the PCA-SEL, a multi-data-driven fusion method for estimating SOH is proposed, called PCA-SEL-FM. The proposed method combines the performance advantages of multi-machine learning algorithms. In the PCA-SEL-FM, GPR, SVR, LSTM, and ELM are selected as four preferred base-learners and ELM as the meta-learner by numerical experiments.
(4) The effectiveness and generality of PCA-SEL-FM is verified by three public datasets, in which the SOH prediction results of PCA-SEL-FM were compared with those of single machine learning algorithms, fusion methods without supplementary feature.

The structure of this paper is as follows: In Section 2, some preparatory tasks are introduced. Section 3 describes the modeling process of the PCA-SEL-FM. Section 4 presents the experimental results and analysis, including the comparison between single ML method, the existing ensemble methods. Meanwhile, the SOH estimation of multiple datasets is presented. Finally, the conclusion is given in Section 5.



(a) two HFs (b) the optimal voltage range of HF2 by TOA

Fig. 2. HF extraction from charging voltage curve of the Cell1.



(a) HF3 (b) the optimal voltage range of HF3 by TOA

Fig. 3. HF extraction from discharging voltage curve of the Cell1.

2. Data analysis

This section outlines several preparatory tasks, including the definition of Li-ion battery SOH, the evaluation indicators, extraction and optimization of multi-dimensional HFs.

2.1. Battery aging experimental dataset

Three publicly available li-ion battery degradation datasets are studied in this paper. Dataset A is provided by the Oxford University which contains four cells (cell1, cell3, cell7, and cell8) [28]. Dataset B consists of three batteries (B0005, B0006, and B0007), which are tested by the NASA Ames Research Center [29]. Dataset C, developed by the Massachusetts Institute of Technology and Stanford University (MIT), focuses on studying the effects of fast charging on battery aging [30]. Two representative li-ion batteries "2017-5-12-21#", and "2018-4-12-1#" are used in this paper and referred to as batch21 and batch1, respectively.

For datasets A and B, the charging process includes constant current (CC) and constant voltage (CV) subprocesses. Initially, the battery is charged at 1C CC until the voltage reaches 4.2 V, then switches to CV mode. The charging process stops when the current decreases to a specified condition, followed by constant discharge at 1C. When the battery terminal voltage reaches a specified cut-off voltage, the discharge process stops. For dataset C, the charging strategy is "C1(Q1)-C2(80%)-1C", where C1 and C2 denote the current of first and second CC steps, respectively, and Q1 is the state of charge when the current switches. The battery remains in CC phase until SOH reaches 80 %, then charges at 1C with CC-CV and discharges at 4C. Table 1 lists the test parameters of three battery datasets and the degradation curve of Li-ion batteries is plotted in Fig. 1.

2.2. SOH definitions and evaluation indicators

SOH serves as an essential indicator for assessing battery performance degradation, which is quantified by factors such as capacity and resistance. In this paper, SOH is specifically defined as the ratio of current battery capacity to nominal capacity:

$$SOH = \frac{Cap_{now}}{Cap_{nom}}, \tag{1}$$

where Cap_{now} is the battery capacity at the current moment, Cap_{nom} is the nominal capacity.

In this paper, root mean square error (RMSE) and mean absolute percentage error (MAPE) are used to evaluate the prediction method's performance.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left|SOH_i - S\widehat{OH}_i\right|}, \tag{2}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{SOH_i - S\widehat{OH}_i}{SOH_i}\right| \times 100\%, \tag{3}$$

where N represents the number of samples in the test set, SOH_i is the true value of SOH in the ith cycle, and $S\widehat{OH}_i$ is the predicted SOH.

2.3. Multi-dimensional HFs extraction and optimization

During li-ion battery operation, key operational data such as voltage and current are captured, providing significant information regarding battery degradation. And multi-dimensional HFs are extracted from these operational data.

1) Charging voltage

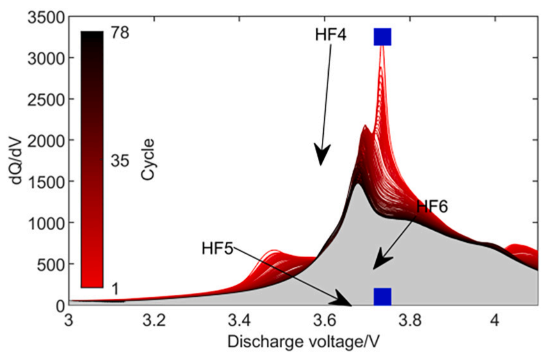


Fig. 4. The smoothing IC curves of discharge cycles.

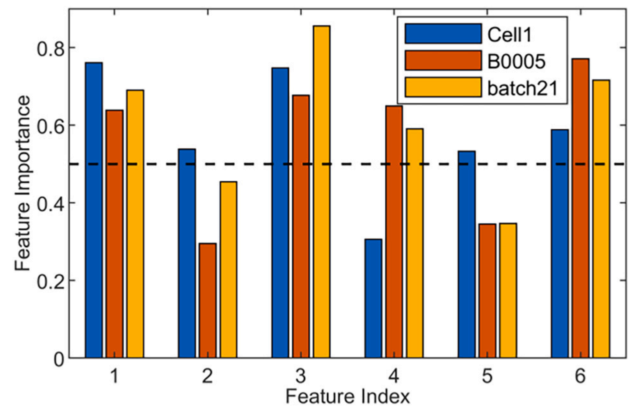


Fig. 5. The importance scores of the six features.

Table 2
Feature parameters of the three datasets.

Number	Description
HF1	The charging time under constant current
HF2	The slope of charging voltage curve
HF3	The time of equal discharge voltage drop
HF4	The peak value of IC curve
HF5	The corresponding voltage of the IC peak
HF6	The area under the IC curve

Table 4
Optimal feature.

Dataset	optimal feature
A	HF1, HF3, HF6
B	HF3, HF6
C	HF1, HF3, HF6

As the cycle progresses, the electrolyte and electrode materials within battery are constantly consumed, leading to a degradation trend in the charging voltage curve. And the charging time under constant current [31], and the slope of charging voltage curve [32] have been confirmed as valuable HFs for predicting SOH. For the slope of charging voltage curve, the effectiveness is significantly influenced by the voltage range. To extract the optimal voltage range, a traversal optimization algorithm (TOA) [33] is introduced. The range of voltage traversal is set as [3.0 V, 4.2 V] and the traversal step size is set as 0.1 V. The optimal voltage result of Cell1 is shown in Fig. 2(b). Thus, the optimal charging voltage intervals of Oxford dataset, MIT dataset and NASA dataset are determined to be [3.9 V, 4.2 V], [3.3 V, 3.6 V] and [3.8 V, 4.1 V], respectively.

2) Discharging voltage

As shown in Fig. 3, when the battery discharges, the voltage curve is steeper in the beginning and end stage, while the intermediate state curve is flatter, can be called the voltage plateau. Meanwhile the time of voltage plateau becomes shorter as the cycle increases, which is called the time of equal discharge voltage drop (TEDVD). Similarly, the TOA algorithm is used to realize the optimal voltage interval selection of TEDVD. And the optimal charging voltage intervals of Oxford dataset, MIT dataset and NASA dataset are determined to be [3.9 V, 3.3 V], [3.2 V, 2.5 V] and [4.1 V, 2.7 V], respectively.

3) Incremental capacity curve of discharge process

The incremental capacity analysis is also an important idea to study voltage curves. A flatter voltage plateau can be transformed into an IC curve with distinct characteristic variations. And the peak value of IC curve [34], the corresponding voltage of the IC peak [35], and the area

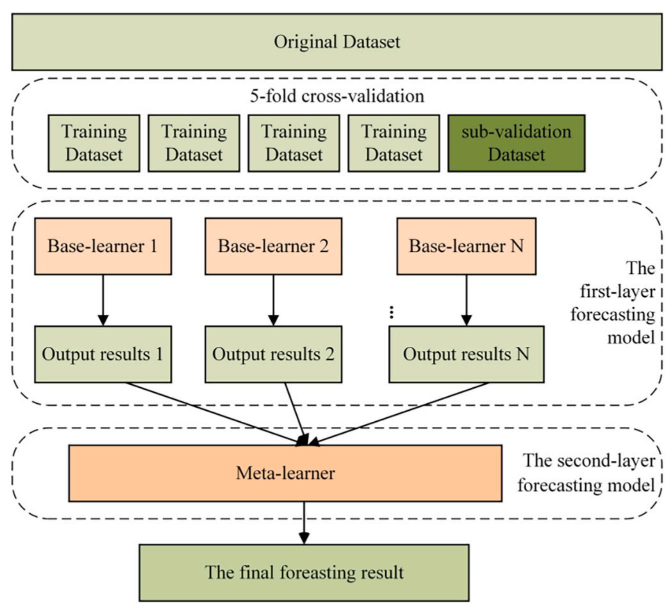


Fig. 6. The structure of Stacking ensemble learning.

Table 3
Pearson correlation coefficient between extracted HFs and SOH.

	Battery	HF1	HF2	HF3	HF4	HF5	HF6
A	Cell1	0.9998	−0.9983	0.9999	0.9444	0.9591	−0.9997
	Cell3	0.9998	−0.9980	0.9998	0.9375	0.9470	−0.9998
	Cell7	0.9998	−0.9982	0.9998	0.9269	0.9290	−0.9997
	Cell8	0.9998	−0.9979	0.9998	0.9378	0.9223	−0.9997
B	B0005	0.9960	−0.9299	0.9962	0.9910	0.9459	0.9945
	B0006	0.9905	−0.8076	0.9949	0.9800	0.9832	0.9995
	B0007	0.9913	−0.8310	0.9978	0.9251	0.9482	0.9963
C	batch21	−0.7924	−0.9675	0.9974	0.9784	0.9505	−0.9882
	batch1	−0.7066	−0.9941	0.9885	0.9736	0.9790	−0.9868

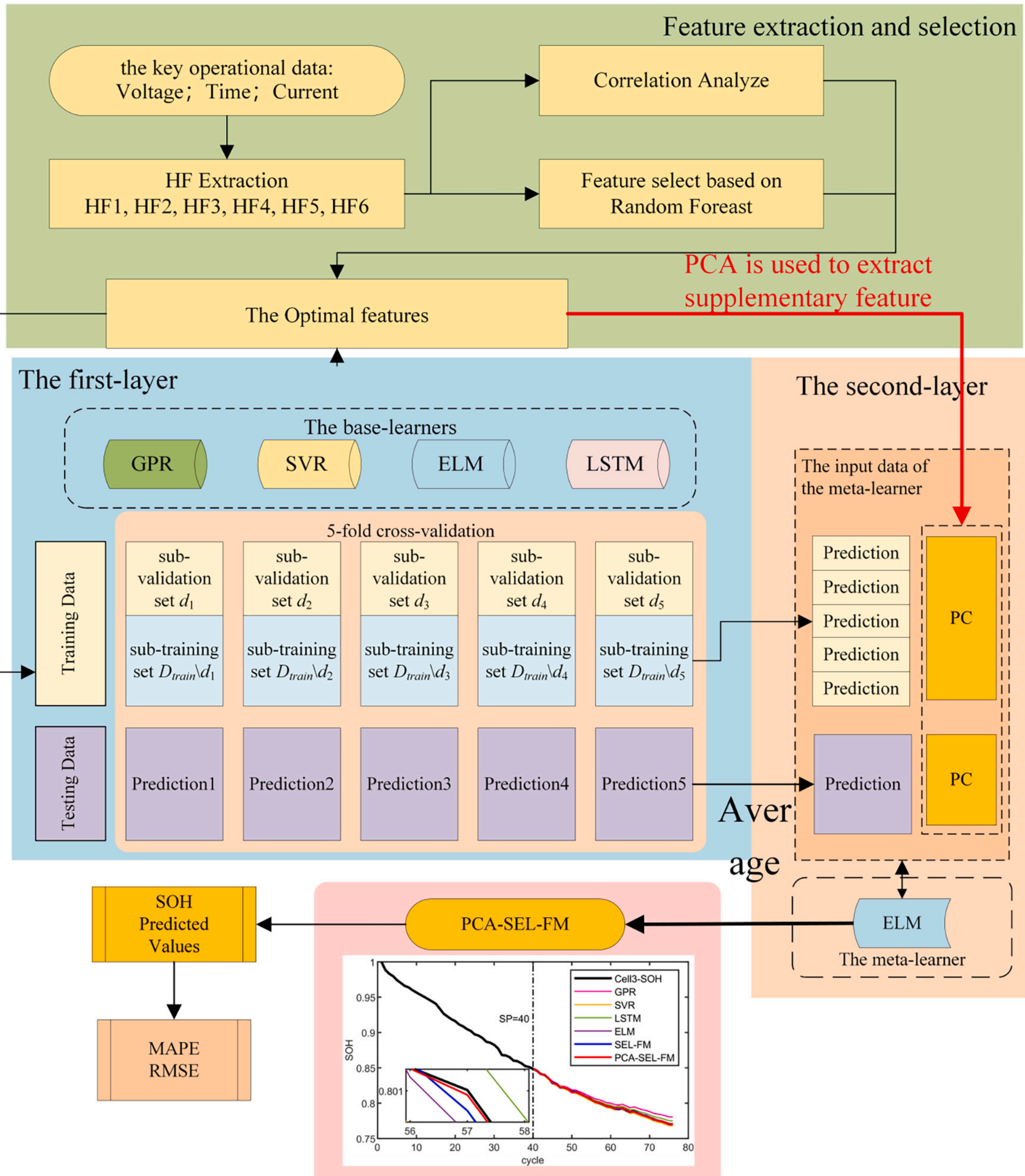


Fig. 7. The overall framework of the PCA-SEL-FM.

under the IC curve [36] have also demonstrated their efficacy as health features for SOH estimation. And the Savitzty-Golay (SG) filtering method [34] is introduced to eliminate the influence of measurement noise and obtain a smooth IC curve. Fig. 4 shows the smoothing IC curve of the Cell1.

In this paper, six feature parameters related to SOH are extracted from the charging and discharging voltage data, which are summarized in Table 2.

To quantitatively analyze the correlation of extracted HFs and SOH, the Pearson correlation coefficient (PCC) is introduced. The PCC can quantitatively reflect the linear correlation between two random variables and can be calculated using the following Eq. (4):

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - E^2(X)}\sqrt{E(Y^2) - E^2(Y)}}, \tag{4}$$

where X is SOH, and Y is the extracted HF.

The PCC results between six HFs and SOH are listed in Table 3. It can be seen that the PPC value are all greater than 0.8, indicating a strong correlation. Due to the influence of fast-charging, the HF1 of two batteries of MIT dataset has low correlation and will be removed in subsequent experiments.

Although each feature has a high correlation with SOH, six HFs are extracted from the voltage data recorded, which may contain redundant information. Training with a dataset containing redundant features increases the complexity of the learning task. The utilization of feature selection methods can effectively reduce information redundancy, thereby enabling the learning algorithm to train more efficient models and improve prediction accuracy. The Random Forest (RF) [37] can randomly select M features from the all-feature set and rank them in terms of importance to evaluate the feature variables. In this paper, the number of decision trees is set to 100, and each tree has a leaf node count of 5. The importance scores of six features are shown in Fig. 5. For Cell1, the scores for the six features are 0.74, 0.57, 0.77, 0.28, 0.56, and 0.53

respectively. For B0005, the top three scores are HF6, HF3, and HF2. For batch21, the features with scores above 0.5 are HF1, HF3, HF4, and HF6.

In summary, based on the analysis results of correlation coefficients and the importance scores, the most suitable input features of each battery dataset are matched to avoid the increase of learning task complexity caused by redundant information. There are listed in Table 4.

3. A heterogeneous learner fusion method with supplementary feature for SOH estimation

Firstly, the traditional stacking ensemble learning method is briefly introduced. To solve the problem that the meta-learner does not directly interact with the original input data in traditional stacking ensemble learning, an improved stacking ensemble learning algorithm based on PCA supplementary feature algorithm is proposed in detail, called PCA-SEL. Finally, a prediction framework for battery SOH based on PCA-SEL is proposed, realizing the integration of multiple heterogeneous learners.

3.1. Stacking ensemble learning

Stacking ensemble learning is a structure that integrates multiple ML algorithms to enhance overall performance, which includes some base-learners and a meta-learner [23]. These base-learners are diverse ML algorithms aimed at extracting information from the original data from different perspectives. The meta-learner integrates the outputs of base-learners to produce the final model results. In addition, to effectively mine meaningful information from limited samples, a 5-fold cross-validation method is commonly employed during the training of base-learners. Fig. 6 illustrates the structure of Stacking ensemble learning algorithm.

Firstly, the original data set is divided into five subsets of equal size using the 5-fold cross-validation method, one subset is used as the validation set, while the remaining subsets are used for training model. Each subset is fed into different base-learners, generating individual prediction results. Subsequently, these results from multiple base-learners are aggregated into a new dataset by averaging to create inputs of the meta-learner. Finally, the meta-learner is trained on the integrated dataset to produce the final prediction result.

3.2. An improved Stacking ensemble learning algorithm-PCA-SEL

From the Fig. 6, it can be seen that the meta-learner of the traditional stacking ensemble learning does not have access to the original data directly. While the design offers certain benefits, it can potentially lead to the loss of crucial information from the original data, thereby affecting the final predictive capabilities. One way to address the issue is by incorporating raw data as additional features in the input for the meta-learner. However, this approach can result in a sudden increase in the input data dimension, potentially introducing redundant information and subsequently escalating the computational complexity of the model.

Principal Component Analysis (PCA) offers a solution by transforming high-dimensional data into a lower-dimensional representation while preserving most of the original information and reducing redundancy. Specifically, the PCA-processed principal components (PC) (those with a contribution rate exceeding 90 %) can be used as supplementary input for the meta-learner, enabling it to better capture essential information from the original input data and optimize the final estimation results.

Thus, an improved stacking ensemble learning algorithm with supplementary feature is proposed, called PCA-SEL. The main flow of PCA-SEL algorithm is shown in Algorithm 1.

3.3. A SOH estimation fusion method based on PCA-SEL

In this paper, the PCA-SEL algorithm is utilized for hierarchical fusion of various single heterogeneous learners. Combining stacking ensemble learning with PCA supplementary feature, a new multi-heterogeneous learners fusion method is proposed for estimating the SOH of li-ion batteries, named PCA-SEL-FM. The details are as follows:

(1) Multi-dimensional HFs extraction and optimization: six HFs are extracted and from the voltage curve of li-ion battery. And three optimal features are selected to generate the original dataset $D = \{(X_i, SOH_i)\}$, where $X_i = [HF_i^1, HF_i^2, HF_i^3], i = 1, \cdots, N$.

Al-
gorithm 1PCA-SEL.

Input: Dataset $D = \{(X_i, Y_i)\}$, $i = 1, \cdots, N$, base-learner $h_{\mathrm{bl}}(\cdot)$, $\mathrm{bl} = 1, \cdots, T$, meta-learner h_{ml}.
Output: Final prediction results $H' = \left\{\hat{Y}_i, i = 1, \cdots, N\right\}$.

Step1: Calculate the principal component features $PC = \{PC_1, PC_2, \cdots, PC_N\}$ as the supplementary input for the meta-learner.
Step2: Use 5-fold cross-validation to generate training dataset and train base-learners:
Step2.1: Use 5-fold cross-validation to generate training.
Randomly divide five sets of equal size: dataset $D = \{d_1, d_2, d_3, d_4, d_5\}$, while d_k is selected as the validation set and the rest $D \setminus d_k$ as the training set.
Step2.2: Train base-learners.

```
for k to 1, ..., 5 do
    for bl to 1, ..., T do
        Train the base-learner h_bl(·) based on the dataset D \ d_k;
    end
    for X ∈ d_k do
        Get the prediction result of the validation set ^k_bl Y = h_b1(X);
    end
end
```

Step3: Use the supplementary features and the prediction results of base-learners to generate a new dataset for meta-learner $D' = \{X', Y\}$, $X' = \left[PC, \left(\sum_{k=1}^{5} {}_{\mathrm{bl}}^{k}Y\right)/5\right]$.
Step4: Train the meta learner to obtain the final prediction $H' = \hat{Y}_i = h_{\mathrm{ml}}(X')$, return H'.

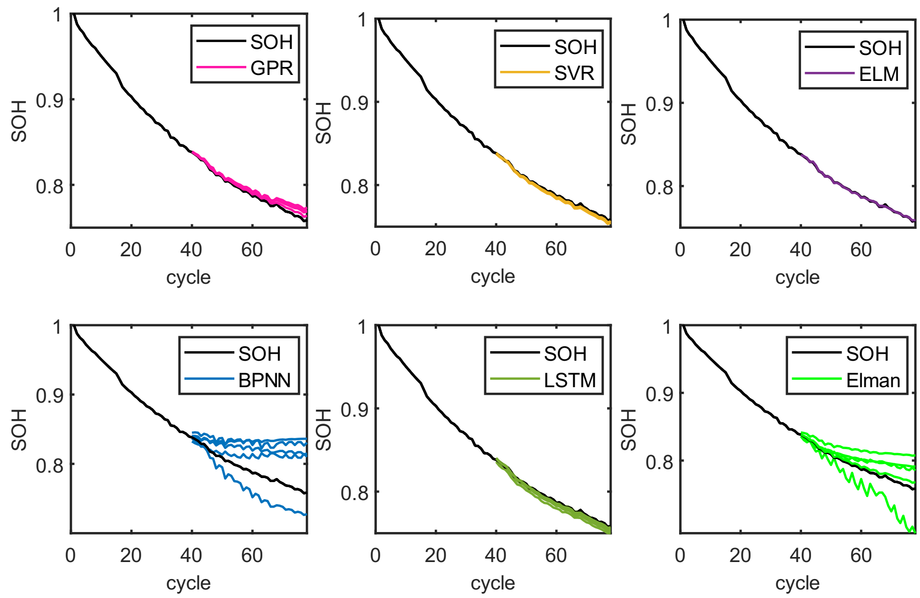


Fig. 8. The SOH estimation of six alternative base-learners after 5-fold cross-validation.

Table 5
The RMSE of each alternative based-learners after 5-fold cross-validation.

Fold	GPR	SVR	ELM	LSTM	BPNN	ELMAN
1	0.0053	0.0011	0.000407	0.0026	0.0287	0.0173
2	0.0073	0.0033	0.000401	0.0075	0.0473	0.0158
3	0.0059	0.0028	0.000482	0.0042	0.0351	0.0321
4	0.0054	0.0016	0.000541	0.0010	0.0268	0.0291
5	0.0028	0.0018	0.000681	0.0016	0.0445	0.0047
Average	**0.0053**	**0.0021**	**0.000502**	**0.0034**	0.0365	0.0198

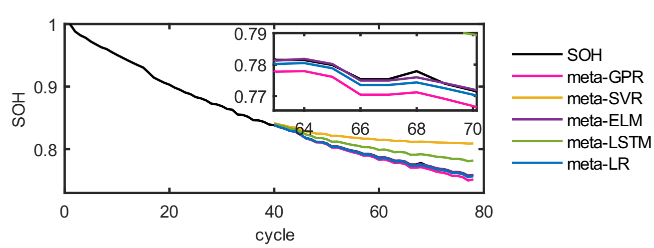


Fig. 9. The SOH estimation of four meta-learners.

(2) Supplementary feature extraction based on PCA: For D, calculate the principal component feature $PC = \{PC_1, PC_2, \cdots, PC_N\}, PC_i = PCA(X_i)$ with a contribution rate greater than 90 % as supplementary feature of the meta-learner.
(3) Divide the original dataset: select a prediction starting point (SP) and generate the initial training set and the initial testing set:$D_{\text{Train}} = \{(X_i, SOH_i), i = 1, \cdots, SP\}$, $D_{\text{Test}} = \{(X_i, SOH_i), i = SP, \cdots, N\}$..
(4) Train base-learners: The initial training set is randomly divided into five subsets of similar size using the 5-fold cross validation method, $D_{Train} = \{d_1, d_2, d_3, d_4, d_5\}$. One subset is used as the sub-training set $D_{Train} \backslash d_k$, while the remaining part serves as the sub-validation set $d_k (k = 1, \cdots, 5)$.

For the k-th fold $(k = 1, \cdots, 5)$, taking base-learner $\text{bl}(k = 1, \ldots, T)$ as an example, train the base-learner $h_{\text{bl}}(\cdot)$based on the sub-training set $D_{Train} \backslash d_k$.

For each sample in the sub-validation set $X_i \in d_k$, the SOH estimation values of the base-learner $h_{\text{bl}}(\cdot)$ can be determined as ${}^{\text{bl}}y_i = h_{\text{bl}}(X_i), i = 1, \cdots, SP$.

Table 6
The estimation results of alternative meta-learners.

	GPR	SVR	ELM	LSTM	LR
RMSE	0.0039	0.0298	**4.34e-04**	0.0134	0.0013

For each sample in the initial training set $X_i \in D_{Test}$, the prediction values of the base-learner $h_{\text{bl}}(\cdot)$ can be obtained as ${}^{\text{bl}}_{k}y_i$, ${}^{\text{bl}}_{k}y_i = h_{\text{bl}}(X_i)$, $\text{bl} = 1, \cdots, T$, $i = \text{SP} + 1, \cdots, N$. Repeat the training until $k = 5$ to end.

(5) Construct the training set and testing set of the meta-learner: The prediction results of based-learners and the PCA supplementary feature is integrated into the training set of the meta-learner:

$D'_{Train} = \{(x_i, SOH_i)\}$, where $x_i = \left[PC_i, {}^{\text{bl}}y_i\right], i = 1, \cdots, \text{SP}$. The testing set of the meta-learner is $D'_{Test} = \{(x_i, SOH_i)\}$, where $x_i = \left[PC_i, \left(\sum_{k=1}^{5} {}^{\text{bl}}_{k}y_i\right)/5\right], i = \text{SP} + 1, \cdots, N$.

(6) Train the meta-learner to obtain the final SOH estimation: Train the meta-learner $h_{\text{ml}}(\cdot)$ based on the dataset D'_{Train} to obtain the final SOH estimation $\widehat{SOH}_i = h_{ml}(x_i), i = \text{SP} + 1, \cdots, N$.

In this paper, ELM as the meta-learner is employed to combine four heterogeneous learners (GPR, SVR, ELM, LSTM) based on the PCA-SEL-FM. The rationale for choosing these specific learners will be discussed in Section 4.1. The overall framework of predicting SOH of li-ion batteries using the fusion method PCA-SEL-FM is shown in Fig. 7.

4. SOH prediction results and analysis

Firstly, the reason behind selecting GPR, SVR, LSTM, and ELM as four preferred base-learners of the fusion method PCA-SEL-FM, and ELM as the meta-learner, is explained. Then, an experiment comparing the fusion method PCA-SEL-FM with individual ML methods (GPR, SVR, LSTM, ELM) using the Oxford dataset is conducted to demonstrate the effectiveness of stacking ensemble learning. Additionally, a fusion method without PCA supplementary feature based on Stacking ensemble

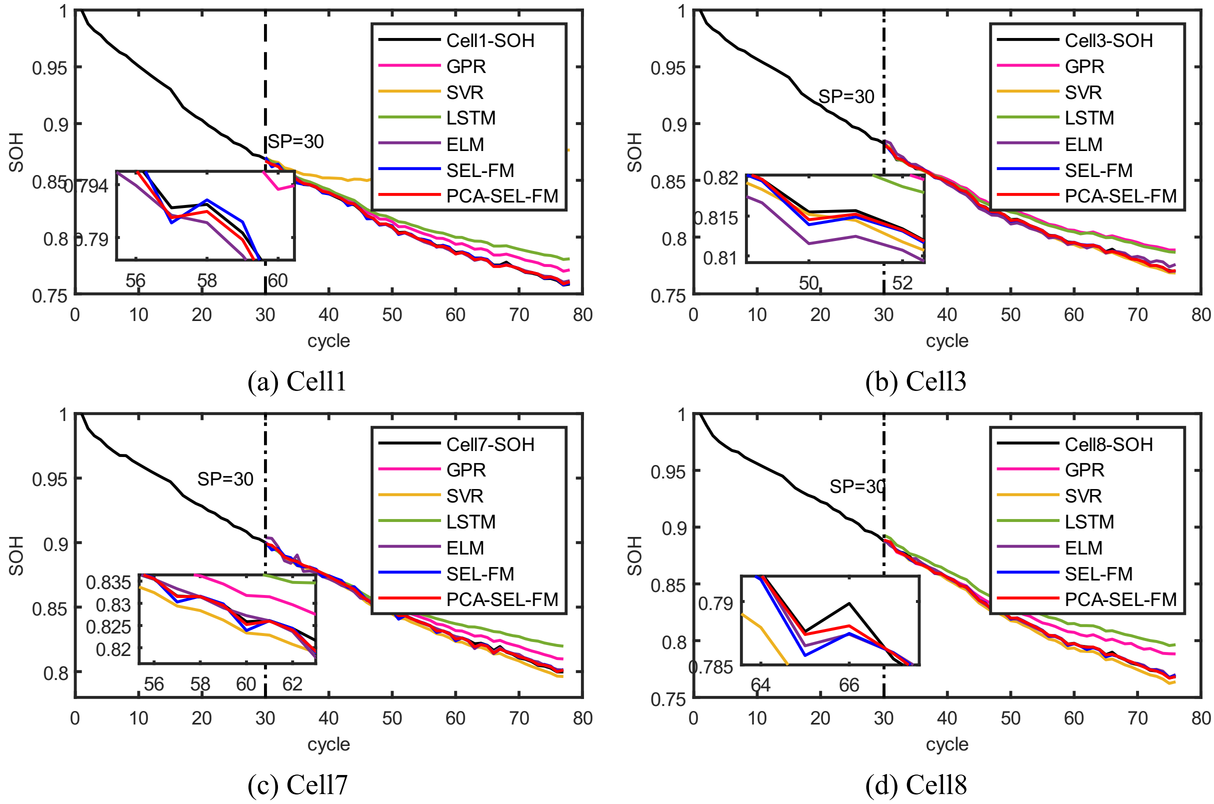


Fig. 10. When SP = 30, SOH prediction results of Oxford dataset based on the PCA-SEL-FM.

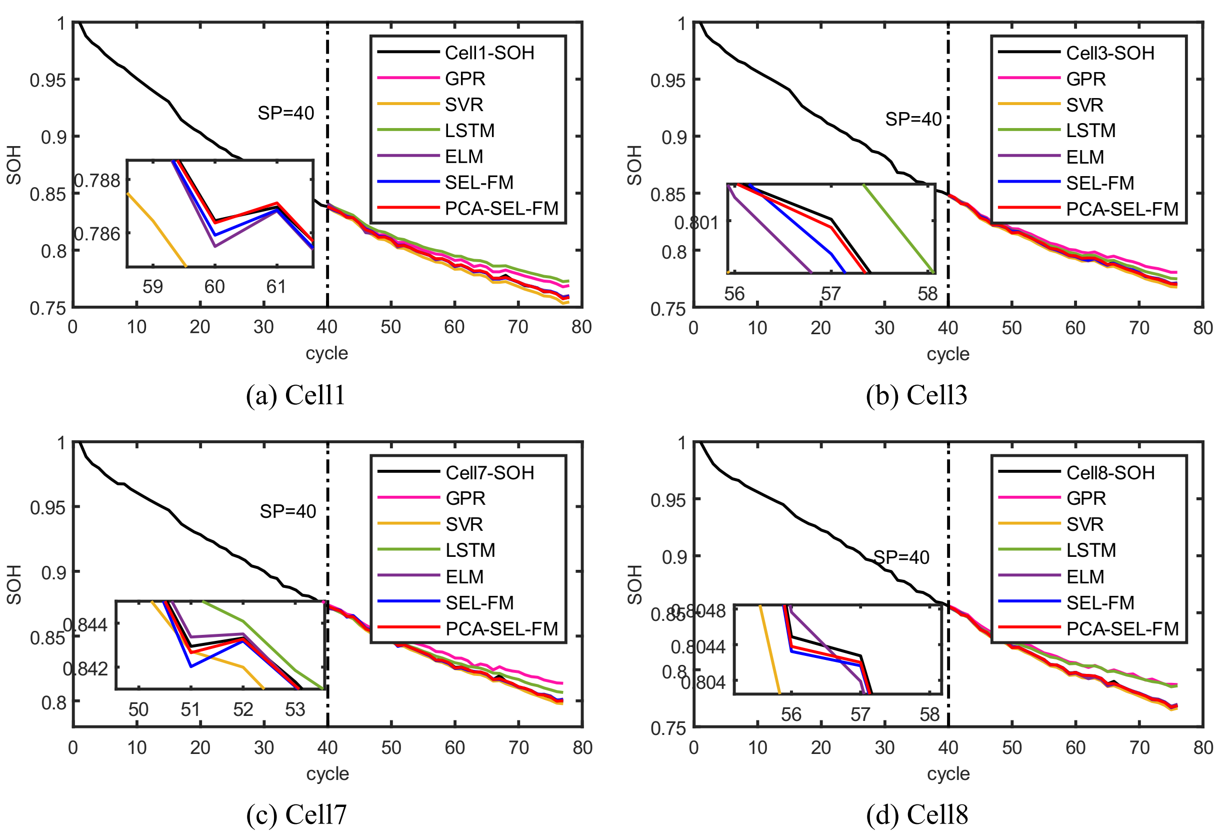


Fig. 11. When SP = 40, SOH prediction results of Oxford dataset based on the PCA-SEL-FM.

Table 7
Estimated results of the Oxford dataset.

SP	Battery	Method	MAPE (%)	RMSE
30	Cell1	GPR	0.73	0.0068
		SVR	6.98	0.066
		LSTM	1.31	0.0121
		ELM	0.13	0.0012
		SEL-FM	0.12	0.0011
		PCA-SEL-FM	**9.49e-02**	**9.26e-04**
	Cell3	GPR	1.03	0.0099
		SVR	0.15	0.0013
		LSTM	0.89	0.0088
		ELM	0.30	0.0029
		SEL-FM	6.95e-02	7.10e-04
		PCA-SEL-FM	**5.39e-04**	**6.25e-04**
	Cell7	GPR	0.51	0.0052
		SVR	0.29	0.0027
		LSTM	0.94	0.0099
		ELM	0.19	0.0022
		SEL-FM	0.11	0.0012
		PCA-SEL-FM	**8.10e-02**	**8.92e-04**
	Cell8	GPR	0.96	0.0098
		SVR	0.38	0.0034
		LSTM	1.76	0.0159
		ELM	0.12	0.0012
		SEL-FM	8.11e-04	8.68e-04
		PCA-SEL-FM	**4.86e-02**	**5.57e-04**
40	Cell1	GPR	0.73	0.0053
		SVR	6.98	0.0034
		LSTM	1.31	0.0085
		ELM	0.10	9.90e-04
		SEL-FM	5.29e-02	5.85e-04
		PCA-SEL-FM	**2.95e-02**	**3.87e-04**
	Cell3	GPR	0.15	0.0057
		SVR	0.45	0.0021
		LSTM	0.72	0.0027
		ELM	0.08	0.0011
		SEL-FM	5.06e-02	6.03e-04
		PCA-SEL-FM	**3.15e-02**	**3.80e-04**
	Cell7	GPR	0.81	0.0074
		SVR	0.78	0.0016
		LSTM	0.32	0.0034
		ELM	8.33e-02	9.14e-04
		SEL-FM	7.01e-02	7.67e-04
		PCA-SEL-FM	**3.26e-02**	**4.61e-04**
	Cell8	GPR	1.83	0.0101
		SVR	0.08	0.0018
		LSTM	0.49	0.0092
		ELM	0.01	0.0010
		SEL-FM	3.66e-02	4.50e-04
		PCA-SEL-FM	**2.96e-02**	**4.10e-04**

(called SEL-FM) is designed to emphasize the importance of supplementary feature for the meta-learner. Finally, the NASA dataset is used to compare the proposed fusion method PCA-SEL-FM and existing ensemble learning methods for SOH estimation, highlighting the superiority of the PCA-SEL-FM.

4.1. The optimal selection of base-learners and meta-learner

Various machine learning algorithms such as GPR, SVR, ELM, LSTM, back propagation neural network (BPNN), and ELMAN neural network are frequently utilized in SOH estimation. However, utilizing too many base-learners can lead to increased computational complexity. Therefore, it is essential to conduct experiments to determine the most suitable base learner for the task. Here, these are selected as potential base-learners to select preferred ones.

For Cell1, we set the SP as 40. Following 5-fold cross-validation, the RMSE and predicted results for six base-learners on the original testing dataset are presented in Fig. 8 and Table 5, respectively.

Fig. 8 shows that the estimated curves of the first four learners are closer to the true curve. From Table 5, it is evident that the average RMSE of GPR, SVR, ELM, and LSTM on the testing set is smaller. However, BP and ELMAN show poor estimation effects due to the randomness of weight matrix selection. Therefore, GPR, SVR, ELM, and LSTM are selected as four preferred base-learners of the fusion method PCA-SEL-FM.

In the past five years, ensemble learning methods have been utilized for SOH estimation. GPR, SVR, LSTM, ELM, and Linear Regression (LR) are be selected as meta-learners. Based on four preferred base-learners mentioned, a fusion method for SOH estimation is developed using PCA-SEL algorithm and various meta-learners. The SOH estimation results of alternate meta-learners on the initial test set are shown in Fig. 9 and Table 6.

It can be seen from Fig. 9 that the purple curve (representing the meta-learner ELM) closely aligns with the original degradation curve, followed by the blue curve (representing the meta-learner LR), while the yellow curve (representing the meta-learner SVR) significantly deviates from the true curve. This indicates that when ELM is used as the meta-learner, the estimated curve of the fusion method PCA-SEL-FM best matches the true degradation curve. The results in Table 6 further support the observation: the RMSE on the test set is smallest when ELM is the meta-learner, at 3.87e-04; LR follows with 0.0013. LR is computationally simple but lacks predictive power for nonlinear problems; on the other hand, ELM offers rapid learning and robust nonlinear processing capabilities, making it more suitable for addressing nonlinear degradation issues in li-ion batteries. As a result, ELM is chosen as the preferred meta-learner for the PCA-SEL-FM.

4.2. SOH estimation based on the Oxford dataset

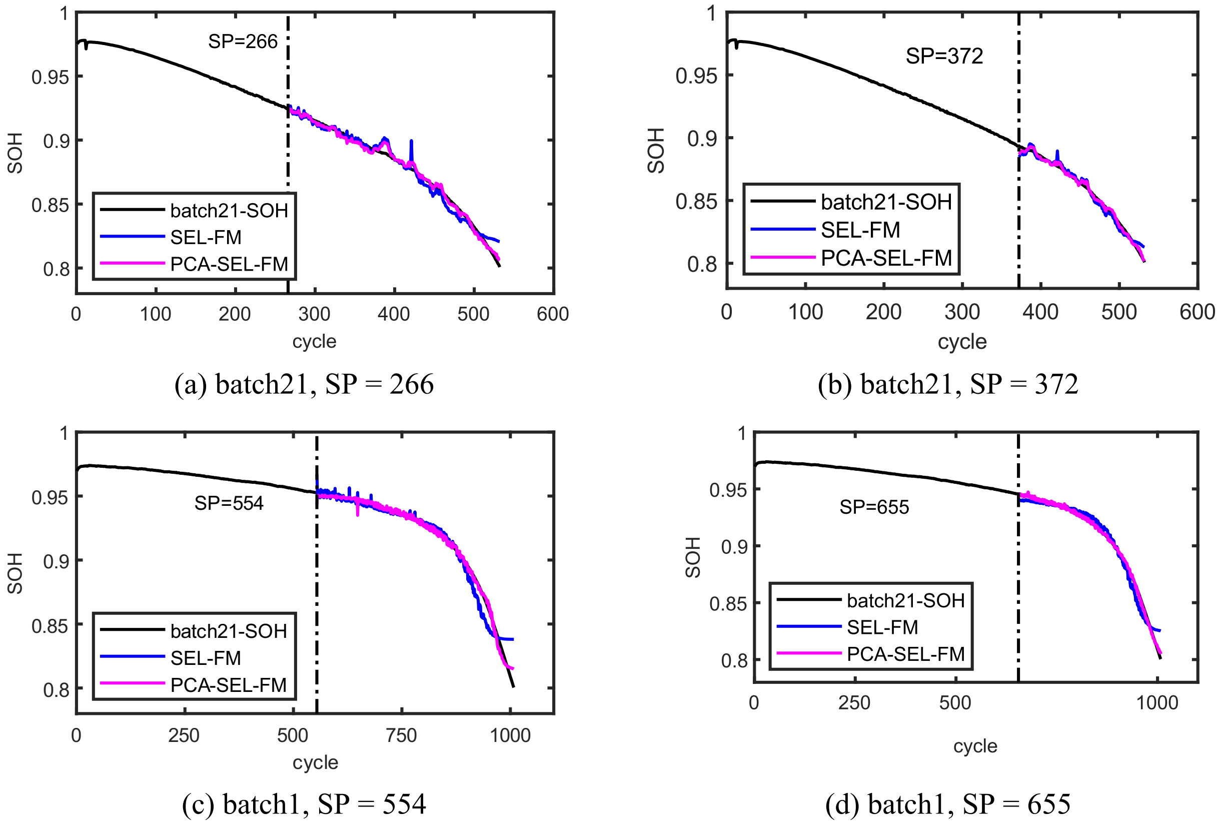
In this section, experiments are conducted on four batteries in the Oxford datasets. HF1, HF3, and HF6 are used as inputs, SOH is used as the output of the PCA-SEL-FM. In addition, we design experiments with different prediction SP, specifically set to 30 and 40. To further verify the accuracy of PCA-SEL-FM, four single machine learning methods (GPR, SVR, LSTM and ELM) and the fusion method without supplementary feature SEL-FM are chosen for comparison. The estimated results are summarized as shown in Fig. 10, Fig. 11, and Table 7.

Fig. 10 and Fig. 11 show the SOH prediction results for four batteries. It can be seen that all six methods demonstrate the ability to effectively track battery degradation trend. Notably, the SEL-FM and PCA-SEL-FM stand out in reducing estimation errors by integrating the results of four individual machine learning methods, highlighting the effectiveness of the stacking ensemble learning structure. Furthermore, it is evident that the red curve of PCA-SEL-FM aligns closely with the actual black curve in comparison to the blue curve of SEL-FM, indicating a higher level of accuracy in estimation. This enhanced accuracy can be attributed to PCA-SEL-FM's capability to uncover additional hidden information through the incorporation of supplementary features into the meta-learner, resulting in more precise predictions.

From Table 7, compared with four single machine learning methods and the SEL-FM, the PCA-SEL-FM has the smallest error, with a RMSE of less than 0.001. In four individual methods, ELM shows the highest accuracy in standalone predictions, thus reinforcing the validity of ELM as a meta-learner. Moreover, there is a difference of approximately 0.025 % in MAPE between PCA-SEL-FM and SEL-FM, indicating that PCA can offer supplementary degradational information to the meta-learner for enhanced accuracy.

For Cell1, when the SP increases from 30 to 40, the MAPE of the PCA-SEL-FM decrease from 0.0949 % to 0.0295 % and RMSE drops from 9.26e-04 to 3.87e-04. Furthermore, with an increase in training set samples, the SOH estimation performance of each method shows gradual improvement, aligning with fundamental principles.

In summary, the PCA-SEL-FM not only fuses multiple heterogeneous models through stacking ensemble learning, but also uses supplementary feature to provide additional degradation information for the meta-learner, achieving more accurate estimation.



(a) batch21, SP = 266 (b) batch21, SP = 372

(c) batch1, SP = 554 (d) batch1, SP = 655

Fig. 12. SOH prediction results for MIT dataset.

Table 8
Estimated results of the MIT dataset.

Batch	SP	Method	MAPE (%)	RMSE
batch21	266	SEL-FM	0.44	0.0054
		PCA-SEL-FM	0.23	0.0027
	372	SEL-FM	0.38	0.0040
		PCA-SEL-FM	0.19	0.0020
batch1	554	SEL-FM	0.55	0.0082
		PCA-SEL-FM	0.20	0.0027
	655	SEL-FM	0.51	0.0061
		PCA-SEL-FM	0.12	0.0013

4.3. SOH estimation based on the MIT dataset

Compared with the Oxford data set, the battery cycle in the MIT data set ranges from 550 to 1000, which is more in line with the long-life characteristics of li-ion batteries. To further validate the effectiveness of the PCA-SEL-FM, similar experiments are carried out using the MIT dataset. Two HFs related to battery degradation are extracted from the charge and discharge voltage curves to serve as inputs for the PCA-SEL-FM, with SOH as the corresponding output. For the PCA-SEL-FM, GPR, LSTM, and ELM are utilized as the preferred base-learners, with ELM serving as the meta-learner. Experiments are conducted with different prediction SP, and the SOH prediction results are shown in Fig. 12. As Fig. 12 clearly shows, the PCA-SEL-FM demonstrates improved accuracy in tracking the degradation of long-cycle li-ion batteries. For batch1, it is evident that the curve predicted by the SEL-FM lacks smoothness in the early stages when SP is set to 554. And the PCA-SEL-FM incorporates principal component feature, enhancing the interaction with the original data and the meta-learner, capturing more degradation information, leading to a smoother estimation curve and a more realistic representation of the SOH estimation curve.

The errors of the MIT dataset are summarized in Table 8. In comparison to the SEL-FM, the PCA-SEL-FM demonstrates lower error rates. For batch21, when SP set as 266, the SEL-FM without PCA supplementary feature yields a MAPE of 0.44 % and RMSE of 0.0054. Conversely, the PCA-SEL-FM, which incorporates PCA supplementary feature, shows a decrease in MAPE by 0.21 % and RMSE by 0.0014. Upon increasing SP to 372, the MAPE of PCA-SEL-FM drops from 0.23 % to 0.19 %. Similarly, for batch1 batteries, PCA-SEL-FM exhibits reduced errors compared to the SEL-FM. All in all, the method PCA-SEL-FM achieves more accurate SOH estimation by adding PCA supplementary features to the meta-learner to supplement the original degradation data. And it is also suitable for SOH estimation of long cycle life batteries.

4.4. Comparison with existing ensemble methods

This section compares the proposed method with existing ensemble learning fusion methods [20,26,38] on the NASA battery dataset. The initial training dataset comprises the first 100 cycles and the remaining cycles from the 101th to 168th are used as testing data to evaluate the performance of the PCA-SEL-FM. In addition, the fusion method without PCA supplementary feature (SEL-FM) is also constructed and the estimation results of SOH of three batteries are shown in Fig. 13.

It can be seen that the SOH estimation curves of SEL-FM and PCA-SEL-FM closely resemble the original degradation curves. However, the relative error in SOH estimation of the PCA-SEL-FM is lower compared to the SEL-FM, and the relative error is controlled within 0.5 %. It indicates that adding PCA supplementary feature to the meta-learner allows for the reuse of information overlooked by base-learners, improving prediction accuracy.

To further validate the efficacy of the PCA-SEL-FM, Table 9 presents a comparison of results with existing fusion methods. The results demonstrate that the RMSE of the PCA-SEL-FM exhibits the lowest value. Compared to previous work [20], the RMSE for three batteries

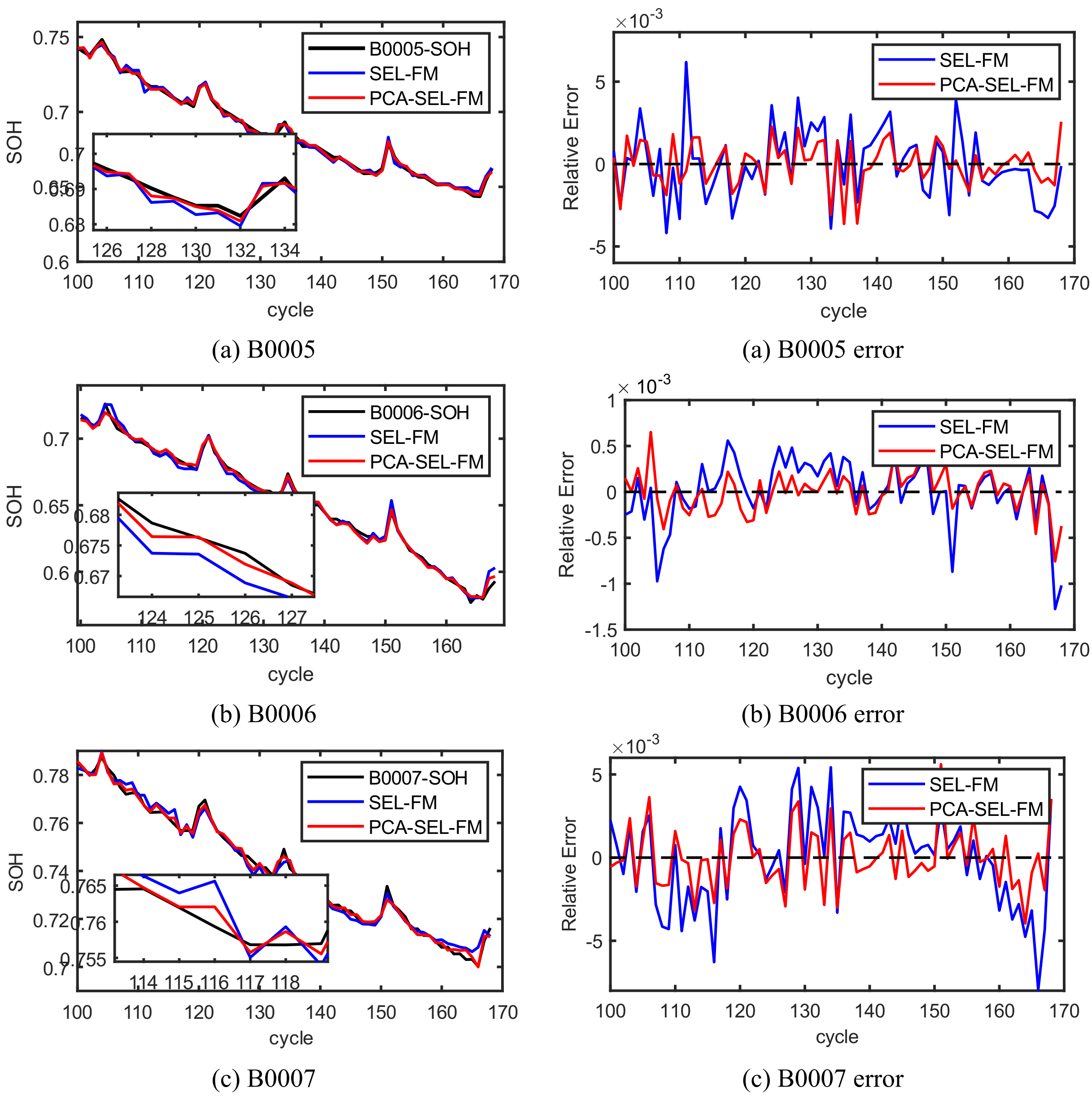


Fig. 13. Estimation results of the PCA-SEL-FM on NASA dataset.

Table 9
The comparison with existing ensemble methods.

	Base-learner	Fusion Strategy	RMSE		
			B0005	B0006	B0007
Zheng, el al. [20]	GPR	Error weight allocation	0.0096	0.0167	00129
Li, el al. [26]	SVR	Stacking LSTM	0.0022	0.0060	0.0032
Lin, el al. [38]	LR, SVR, GPR	Stacking LR	0.0049	0.0129	0.0057
Proposed method	**GPR, SVR LSTM, ELM**	Stacking ELM without supplementary feature	0.0021	0.0035	0.0028
		Stacking ELM with PCA supplementary feature	**0.0013**	**0.0023**	**0.0016**

showed reductions of 0.0083, 0.0154, and 0.0113, respectively and it suggests that the stacking ensemble learning approach yields more precise estimates compared to the error weight allocation strategy. While Li et.al [26] also employs a stacking ensemble learning strategy, the base-learners are solely SVR models, lacking the benefits of heterogeneous integration. Although the advantages of Stacking heterogeneous integration were utilized in another study [38], the choice of LR as the meta-learner posed limitations, known for its limitations in nonlinear modeling. In contrast, we select ELM as the meta-learner, which excels in capturing the nonlinear degradation patterns of li-ion batteries. Moreover, compared to the SEL-FM, the RMSE of the PCA-SEL-FM are lower by 0.0009, 0.0012, and 0.0012, respectively, underscoring the value of incorporating PCA supplementary features into the meta-learner.

5. Conclusion

In this paper, a new heterogeneous learner fusion method based on stacking ensemble learning, PCA-SEL-FM, is proposed for estimating

SOH of li-ion batteries. The method achieves the integration of heterogeneous learners and predicts the SOH of the li-ion battery accurately. The main work is as follows:

Firstly, multi-dimensional health features are constructed and optimized from the charge and discharge voltage data, with the optimal input features identified for the fusion method PCA-SEL-FM. Secondly, an improved stacking ensemble learning algorithm PCA-SEL, is proposed. And PCA is employed to extract principal component features from multi-dimensional health features, which served as supplementary feature for the meta-learner, addressing the issue of limited interaction between the meta-learner and the original data in traditional stacking ensemble learning algorithm. Then, the fusion method PCA-SEL-FM is constructed to realize the online estimation of SOH, realizing the fusion of four heterogeneous learners. Finally, the Oxford and MIT datasets are utilized as experimental validation to demonstrate the excellence of the proposed method and comparative experiments of multiple existing ensemble methods are carried out on NASA datasets. The results show demonstrate that the proposed fusion method not only leverages the strengths of individual methods in SOH estimation but also ensures that the meta-learner can extract more degradation information from li-ion batteries, thereby enhancing the accuracy of SOH estimation.

In future work, we will try to introduce temperature features or adjust fusion strategy to improve the method on the basis of this paper.

CRediT authorship contribution statement

Hailin Feng: Writing – review & editing, Supervision, Formal analysis. **Lu Zhang:** Writing – original draft, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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